Predicate Logic

Not all valid deductions can be formalised in propositional logic.

Example: To express the following logical deduction in propositional logic:

1. All horses are animals.
2. *Therefore,* all horse-heads are animal heads.

we would need to say things like *"if* x *is a horse, then* x *is an animal"*.

In order to do this, we extend propositional logic to include ***predicates***: properties which are true or false of elements of a universe of discourse.

We have actually been using predicates with set-building notation

$$\{\, \mathrm{x} : \mathrm{x} \textit{ has property } \mathrm{P} \,\}$$

The part "x *has property* P" is a predicate, typically denoted $\mathrm{P}(\mathrm{x})$. A predicate is an *indeterminate proposition*, which may be true or false of a particular element.

Predicates and Free Variables

Example:

$$\text{Prime}(\text{x}) \quad = \quad \text{“x is a prime number”}$$

is a predicate which says that x is a prime number. The universe of discourse is probably $\mathbb{N}$; but it could be anything (and $\text{Prime}(\text{x})$ would be false if $\text{x} \notin \mathbb{N}$).

The “x” in “$\text{Prime}(\text{x})$” is called a ***free variable*** as it stands for an unspecified value. If we instantiate x, we would get a proposition. For example

$$\text{Prime}(5) \text{ is true} \quad \text{whereas} \quad \text{Prime}(6) \text{ is false}$$

Example: Given the predicate $\text{Female}(\text{x}) = \text{“x is a female”}$ where the universe of discourse is the Duck family:

$$\text{Ducks} = \{ \text{Quackmore, Hortense, Scrooge, Donald, Della, Huey, Louis, Dewey} \}$$

Then

- $\text{Female}(\text{Hortense})$ and $\text{Female}(\text{Della})$ are true;
- $\text{Female}(\text{Quackmore})$, $\text{Female}(\text{Scrooge})$, $\text{Female}(\text{Donald})$, $\text{Female}(\text{Huey})$, $\text{Female}(\text{Louis})$ and $\text{Female}(\text{Dewey})$ are false.

Truth Sets

The ***truth set*** of a predicate $P(x)$ is the set of objects which satisfy it:

$$\{\, x : \; P(x) \,\}$$

Examples:

The truth set of $\mathrm{Prime}(x)$ is the set of all prime numbers:

$$\{\, 2\,,\, 3\,,\, 5\,,\, 7\,,\, 11\,,\, 13\,,\, 17\,,\, \ldots \,\}.$$

The truth set of the predicate $\mathrm{Female}(x)$ over the Duck family is

$$\{\, \mathrm{Hortense},\ \mathrm{Della} \,\}.$$

The truth set of the predicate *"x is an even integer"* is

$$\{\, \ldots\,,\, -6\,,\, -4\,,\, -2\,,\, 0\,,\, 2\,,\, 4\,,\, 6\,,\, \ldots \,\}.$$

The truth set of the predicate *"x is an even prime number"* is $\{\, 2 \,\}$.

Predicates with two or more free variables

Predicates may range over more than one element.

For example, *equality* ($=$) and *set inclusion* ($\subseteq$) are binary predicates. For these, we typically use ***infix notation*** instead of ***prefix notation***; that is, $x{=}y$ instead of $=(x, y)$, and $A{\subseteq}B$ instead of $\subseteq(A, B)$.

For example: $5=5$ is true; $\emptyset \subseteq \{5\}$ is true; and $\{\emptyset\} = \emptyset$ is false.

The truth set of a predicate which ranges over more than one element consists of a tuple of values.

Example: Let $\mathrm{Divides}\,(x, y)$ be the two-place predicate over integers saying that x *divides evenly into* y. Then

- $\mathrm{Divides}\,(3, 15)$ is true, and $\mathrm{Divides}\,(4, 15)$ is false;
- the truth set of $\mathrm{Divides}\,(x, y)$ is

$$\{\,(x, y) :\ x \text{ divides evenly into } y\,\}$$

(Note: The standard mathematical symbol for $\mathrm{Divides}$ is "$|$", used in infix notation, as in $3 \mid 15$ and $4 \nmid 15$.)

All and Some

We can form a proposition from a (unary) predicate by saying that it is true of *every* element, or that it is true of *some* (i.e., at least one) element.

Example: *"Every prime number greater than 2 is odd"*:

$$\mathrm{Odd}(3) \ \wedge \ \mathrm{Odd}(5) \ \wedge \ \mathrm{Odd}(7) \ \wedge \ \mathrm{Odd}(11) \ \wedge \ \cdots$$

where $\mathrm{Odd}(\mathrm{x})$ says *"*x *is an odd number"*.

Example: *"Some prime numbers are square"*:

$$\mathrm{Square}(2) \ \vee \ \mathrm{Square}(3) \ \vee \ \mathrm{Square}(5) \ \vee \ \cdots$$

where $\mathrm{Square}(\mathrm{x})$ says *"*x *is a square number"*.

These are infinitely-long conjunctions / disjunctions, and cannot be expressed in propositional logic (without using an ellipsis $\cdots$).

Predicate logic provides two forms of *quantification* which will allow to express these in a formula in a finite way: one which expresses that a property holds univerally (always), and another which expresses that a property holds existentially (sometimes).

Universal Quantification

The statement $\forall x\ P(x)$ (pronounced *"for all x, $P(x)$"*) is called ***universal quantification*** and is true if, and only if, $P(x)$ is true of ***all*** possible values of x.

Example: *"Everyone did the homework"* is expressed as:

$$\forall x\, H(x)$$

where $H(x) =$ "x did the homework", and the (assumed) universe of discourse is the set of students who were assigned the homework.

- the variable x in the formula $\forall x\, H(x)$ is called a ***bound variable***; it is *bound* by the quantifier $\forall x$.
- The statement has a definite truth value.

Example: *"Every prime number greater than 2 is odd"* is expressed as:

$$\forall x \left(\text{Prime}(x) \ \wedge\ x > 2 \ \Rightarrow\ \text{Odd}(x) \right)$$

Note: The (outer) parentheses are needed, as universal quantification binds more strongly than all propositional connectives.

Universal Quantification

Example: *"Every dog that has stayed in the kennel will have to go into quarantine"* can be expressed as

$$\forall x \left(K(x) \Rightarrow Q(x) \right)$$

where $K(x) =$ "x *has stayed in the kennel*" and $Q(x) =$ "x *has to go into quarantine*". The (assumed) universe of discourse is the set of all dogs.

Breaking this down:

For all x: if x has stayed in the kennel, then x has to go in to quarantine.

Example: *"Nobody likes a sore loser"* can be expressed as:

$$\forall x \left(S(x) \Rightarrow \forall y \neg L(y, x) \right)$$

where $S(x) =$ "x *is a sore loser*" and $L(y, x) =$ "y *likes* x". The (assumed) universe of discourse is set of all people.

Breaking this down:

For all x: if x is a sore loser, then for all y: y doesn't like x.

Universal Quantification

Exercise: Using the predicates

- $\mathrm{B}(x) =$ "x *is a bee*"
- $\mathrm{F}(x) =$ "x *is a flower*"
- $\mathrm{L}(x, y) =$ "x *likes* y"

write each of the following statements in predicate logic.

1. Bees like all flowers *(For all x: if x is a bee,*
   then for all y: if y is a flower,
   then x likes y)
2. Bees only like flowers *(For all x: if x is a bee,*
   then for all y: if x likes y,
   then y is a flower)
3. Only bees like flowers *(For all x and for all y: if y is a flower*
   and x likes y
   then x is a bee)

Existential Quantification

The statement $\exists x\ P(x)$ (pronounced *"there exists* x *such that* $P(x)$*"*) is called ***existential quantification*** and is true if, and only if, $P(x)$ is true of ***some*** (at least one) value of x.

Example: *"Some prime numbers are square"* can be expressed as:

$$\exists x\Big(\text{Prime}(x) \wedge \text{Square}(x)\Big)$$

- the variable x in the formula is (again) called a ***bound variable***; it is bound by the quantifier $\exists x$.
- The statement has a definite truth value (it is false).

Example: *"Someone didn't do the homework"* is expressed as:

$$\exists x\ \neg\, H(x)$$

where $H(x) =$ "x *did the homework*", and the (assumed) universe of discourse is the set of students who were assigned to do the homework.

Note: Existential quantification also binds more strongly than all propositional connectives.

Existential Quantification

Example: *"If some dog that has stayed in the kennel has been in contact with a dog with rabies, then every dog that has stayed in the kennel will have to go into quarantine"* can be expressed as

$$\exists x \Big(\mathrm{K}(x) \wedge \exists y \Big(\mathrm{C}(x,y) \wedge \mathrm{R}(y) \Big) \Big) \;\Rightarrow\; \forall x \Big(\mathrm{K}(x) \Rightarrow \mathrm{Q}(x) \Big)$$

where
$\mathrm{K}(x)$ = "x *has stayed in the kennel*"
$\mathrm{R}(x)$ = "x *has rabies*"
$\mathrm{C}(x,y)$ = "x *and* y *have been in contact*"
$\mathrm{Q}(x)$ = "x *has to go into quarantine*"

The (assumed) universe of discourse is the set of all dogs.

Breaking this down:

If there is an x that has stayed in the kennel,
and there is a y that has been in contact with x and has rabies,
then for all x: if x has stayed in the kennel,
then x has to go into quarantine.

Existential Quantification

Exercise: Let $\mathrm{M}(\mathrm{x}, \mathrm{y})$ say *"x is the mother of y"*, where the universe of discourse is the set of all human beings.

Express the following predicates

1. Everyone has a mother

   (For each x *there is a* y *such that*
   y *is the mother of* x*)*

2. Everyone has exactly one mother

   (For each x *there is a* y *such that*
   y *is the mother of* x*, and*
   for every z*: if* z *is the mother of* x*,*
   then $z = \mathrm{y}$*)*

Bounded Quantification

The following are useful restricted forms of quantification which we use for convenience:

- $\forall\, x \in A\; P(x)$ pronounced *"for all* x *in* A, $P(x)$*"*

  is logically equivalent to: $\forall x \left(x \in A \Rightarrow P(x) \right)$

- $\exists\, x \in A\; P(x)$ pronounced *"there exists some* x *in* A *such that* $P(x)$*"*

  is logically equivalent to: $\exists x \left(x \in A \wedge P(x) \right)$

- $\exists!\, x\; P(x)$ pronounced *"there exists a unique* x *such that* $P(x)$*"*

  is logically equivalent to: $\exists x \left(P(x) \wedge \neg\, \exists y\, (P(y) \wedge x \neq y) \right)$

Rules for Quantification

- If it is not the case that $\mathrm{P}(\mathrm{x})$ is true for all x,
  then there must be some x for which $\mathrm{P}(\mathrm{x})$ is false:

$$\neg\forall \mathrm{x}\, \mathrm{P}(\mathrm{x}) \;\Leftrightarrow\; \exists \mathrm{x}\, \neg \mathrm{P}(\mathrm{x})$$

- If it is not the case that $\mathrm{P}(\mathrm{x})$ is true for some x,
  then $\mathrm{P}(\mathrm{x})$ must be false for all x:

$$\neg\exists \mathrm{x}\, \mathrm{P}(\mathrm{x}) \;\Leftrightarrow\; \forall \mathrm{x}\, \neg \mathrm{P}(\mathrm{x})$$

Examples:

$\neg$(Some people like mathematics) $\Leftrightarrow$ Everybody dislikes mathematics

$\neg$(All cats have fur and a tail) $\Leftrightarrow$ Some cats are either bald or tailless

Modelling in Predicate Logic

Exercise: Let *Divides* (k, n) be the statement "k divides evenly into n", where the variables range over the set of positive integers $\mathbb{Z}^+$; that is, $\mathrm{n}{=}\mathrm{dk}$ for some integer d.

Determine the truth value of each of the following statements.

1. *Divides* $(4, 6)$
2. *Divides* $(2, 6)$
3. $\forall \mathrm{k}\, \forall \mathrm{n}$ *Divides* (k, n)
4. $\exists \mathrm{n}\, \forall \mathrm{k}$ *Divides* (k, n)
5. $\exists \mathrm{k}\, \forall \mathrm{n}$ *Divides* (k, n)
6. $\forall \mathrm{n}\, \exists \mathrm{k}$ *Divides* (k, n)

Modelling in Predicate Logic

Example: A Lewis Carroll puzzle deduces that no baby can manage a crocodile from the following assumptions:

- All babies are illogical.
- Nobody is despised who can manage a crocodile
- Illogical persons are despised.

Define the following predicates:

$$\mathrm{B}(x) = \text{“}x \text{ is a baby”} \qquad \mathrm{D}(x) = \text{“}x \text{ is despised”}$$
$$\mathrm{I}(x) = \text{“}x \text{ is illogical”} \qquad \mathrm{M}(x) = \text{“}x \text{ can manage a crocodile”}$$

The three premises above translate into the following:

1. $\forall x \left(\mathrm{B}(x) \Rightarrow \mathrm{I}(x)\right)$ 2. $\forall x \left(\mathrm{M}(x) \Rightarrow \neg \mathrm{D}(x)\right)$ 3. $\forall x \left(\mathrm{I}(x) \Rightarrow \mathrm{D}(x)\right)$

and the conclusion translates to $\forall x \left(\mathrm{B}(x) \Rightarrow \neg \mathrm{M}(x)\right)$.

For any x for which $\mathrm{B}(x)$ is true, we have $\mathrm{I}(x)$ true by 1., then $\mathrm{D}(x)$ true by 3., and finally $\neg \mathrm{M}(x)$ true by 3. Hence, the conclusion indeed follows from the premises.

Modelling in Predicate Logic

Example: Consider the following deduction:

1. All horses are animals.
2. *Therefore,* all horse-heads are animal heads.

Letting

- $\mathrm{Horse}(x) =$ "x is a horse"
- $\mathrm{Animal}(x) =$ "x is an animal" and
- $\mathrm{Head}(x, y) =$ "x is the head of y"

the statements above translate to:

1. $\forall x \Big(\mathrm{Horse}(x) \Rightarrow \mathrm{Animal}(x) \Big)$
2. $\forall x \Big(\exists y (\mathrm{Horse}(y) \wedge \mathrm{Head}(x, y)) \;\Rightarrow\; \exists y \Big(\mathrm{Animal}(y) \wedge \mathrm{Head}(x, y) \Big) \Big)$